# Nicholas Chin Bo You

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## Education

**University of Utah** *August 2024 – Present*
*Salt Lake City, UT*

*BS in Physics with Computational Physics and Astronomy Emphasis*
*BS in Applied Mathematics*

- CGPA: 3.872
- Dean's List Fall 2024, Spring 2025, Fall 2025, Spring 2026, Summer 2026

**Foon Yew High School** *Jan 2018 - Nov 2023*
*Johor Bahru, Malaysia*

## Publications

**Measurement of the photosphere oblateness of $\gamma$ Cassiopeiae via Stellar Intensity Interferometry with the VERITAS Observatory**
*A. Archer, J.P. Aufdenberg, . . . , **N. B. Y. Chin**, . . . , The VERITAS Collaboration*
*The Astrophysical Journal, 995, 191 (2025)*

**Shared Risk-Capital Constraints in Dynamic Option Dealer Hedging**
*Submitted to SIAM Journal on Financial Mathematics*

## Research Experiences

**Analysis of Extended Gamma Ray Sources with VERITAS Using Gammapy**
*University of Utah Gamma Ray Astronomy Research Group | Advisor: Prof. David Kieda | Summer 2026 - Present*

- Investigating two-dimensional spatial and spectral analysis of Cassiopeia A using Gammapy, including morphology and spectral model fitting.
- Implemented a gamma ray extended source analysis pipeline for VERITAS observations.
- *2026 4CS Poster Presentation*

**Verification of VERITAS Results Using the Gammapy Analysis Package (SURF Program)**
*University of Utah Gamma Ray Astronomy Research Group | Advisor: Prof. David Kieda | Summer 2026*

- Implemented a gamma ray data analysis pipeline using Gammapy for VERITAS observations and validated its results through independent comparisons with analyses performed by researchers working on the VERITAS–Gammapy integration.
- Conducted point-source analyses of the Crab Nebula and Cassiopeia A, including background estimation, significance mapping, and spectral modeling to characterize their very-high-energy gamma-ray emission.
- *Summer Symposia Presentation*

**Efficient Quantum State Preparation Algorithms for Astronomical Image Encoding**
*Independent Research | Advisor: Prof. Massood Tabib-Azar | Spring 2026 - Present*

- Investigating efficient quantum state preparation algorithms for encoding astronomical images on quantum computers, by exploiting image sparsity and nonzero structure to reduce quantum circuit complexity.
- Implemented several quantum state-preparation algorithms for Quantum Probability Image Encoding (QPIE), including methods designed to exploit sparsity and structured data.
- Benchmarked the algorithms using circuit complexity metrics and tested the fidelity of the prepared quantum states against their target states.

**Stellar Intensity Interferometry: Search of Modulation Signatures in Variable Star Measurements and Background Analysis**
*VERITAS Stellar Intensity-Interferometry Research Group | Advisor: Prof. Tugdual LeBohec & Prof. David Kieda | Fall 2024 - Spring 2026*

- Refined and maintained Stellar Intensity Interferometry (SII) data analysis workflows using CERN ROOT and C++, improving the processing and interpretation of observational data for the University of Utah SII research group.

- Analyzed observational data from the $\beta$ Cephei variable $\beta$ Canis Majoris to investigate temporal modulation signatures associated with stellar photospheric pulsations.
- Prototyped a data format converter between FPGA-correlated data and ROOT-compatible data structures for integration with the existing analysis framework.
- Performed supporting computations to test and evaluate a moonlight-induced background model for SII observations.
- *2025 4CS Poster Presentation*

## Presentations

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| APS Four Corners Section 2026 Annual Meeting | Fall 2026 |
| University of Utah Undergraduate Research Symposium - Summer Symposia | Summer 2026 |
| Quantum Computing Focused Reading Group | Spring 2026 |
| ◦ Affliated with NSF Research Training Group | |
| APS Four Corners Section 2025 Annual Meeting | Fall 2025 |
| University of Utah Department of Physics and Astronomy Annual Symposium | Spring 2025 |

## Skills

- Languages: English, Mandarin Chinese, Cantonese Chinese, Hakka Chinese, Bahasa Melayu
- Programming Languages: Python, C/C++, Rust, Java, Bash, LaTeX
- Scientific Computing : CERN ROOT, Gammapy, Qiskit, Jupyter Notebook, NumPy, Matplotlib, MATLAB
- Operating Systems: Linux / Unix

## Awards and Certificates

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| Simons Laufer Mathematical Sciences Institute – SACNAS Travel Award | Summer 2026 |
| Summer Undergraduate Research Fellowship (SURF) | Summer 2026 |
| Thomas J. Parmley Scholarship Endowment Fund | Fall 2026 - Spring 2027 |
| MIT iQuHACK 2026 - BlueQubit Problem | World Rank 35, 2nd Highest Score |
| Undergraduate Research Opportunity Program Award (UROP) | Spring 2026 |
| Talent Scholarship | Fall 2025 - Spring 2026 |
| Chen Jingrun's Cup Secondary School Mathematics Competition 2021 | National Rank 81st |
| Chen Jingrun's Cup Secondary School Mathematics Competition 2023 | High Distinction |
| OpenAI Codex Hackathon | 1st runner up |
| IELTS 2024 | 6.5 |
| MATLAB Onramp | Course Completion Certiicate |

## Other Experiences

**Research Assistant** — *U of U*
*University of Utah Astroparticle Physics | Gamma Ray Astronomy Research Group* — *Nov 2024 – Present*

- Working with Prof. Tugdual LeBohec and Prof. David Kieda on Stellar Intensity Interferometry
- Working with Prof. David Kieda on Gamma Ray Astronomy

**Lead Organizer - Qiskit Fall Fest 2026** — *Salt Lake City, UT*
*IBM Quantum* — *Aug 2026 – Present*

- Selected by IBM Quantum from 900+ applicants to lead one of the 200 Qiskit Fall Fest events at the University of Utah.
- Coordinating a team of student organizers and faculty supporters to develop quantum computing workshops, invited talks, community activities, and a student hackathon.

**Chair - Undergraduate Student Advisory Committee** — *U of U*
*U of U Physics Department* — *Oct 2025 – Present*

- Expanded committee membership by 300% through active recruitment and outreach initiatives.
- Organized a department-wide social event in collaboration with the Society of Physics Students (SPS), attracting 30+ participants across undergraduate, grad-

uate, and non-physics programs.

- Led a student town hall attended by 40+ students to collect and communicate departmental feedback to faculty leadership.

**Officer - Blockchain Club** *U of U*
**Treasurer - Fintech Club** *Jan 2025 – Jun 2025*
*U of U Finance Department* *Jul 2025 – Present*

- Developed educational content introducing blockchain, frontier science, and emerging computational technologies.
- Authored educational articles for the club's official website.

**CFO - UCubesat Club** *U of U*
*U of U ECE Department* *Jan 2025 – May 2025*

- Telemetry Team; Ground Station Subsystem Team Lead.
- Working on building a ground station to contribute to the SatNOGS Network.

**Engineering Possibilities Summit Scholar** *Jan 2025 - Dec 2025*
*Goldman Sachs*

**President - Badminton Club** *Sept 2026 - Present*
*U of U Eccles Student Life Center*

**Apprentice Management** *Johor Bahru, Malaysia*
*Rhapsody Youth Ensemble* *2 projects (Approximately 6 months)*

- Established the foundational framework for ensemble's operations and concert planning.

**Co-Chairman and President of Education** *Johor Bahru, Malaysia*
*Foon Yew Symphony Orchestra* *June 2022 - May 2023*

- Utilized recording and mixing skills to support the orchestra during the COVID period, facilitating virtual performances
- Organized and conducted more than 10 hours of online courses on different subjects weekly to maintain member engagement and skill development.
- Designed programs to help students foster interest and performing opportunities. Approximately 80% of members increased 40% more performing opportunities annually

**Assistant Luthier and Sales Assistant** *Johor Bahru, Malaysia*
*Melodia String* *Jan 2024 - Aug 2024*

- Performed walk-in instrument maintenance and basic repairs, including string replacement, setup adjustments, and minor restorations
- Studied and completed projects up to an intermediate level luthiery techniques
- Maintained workshop equipments and tools

**Recording Studio Assistant** *Johor Bahru, Malaysia*
*Goldstone Music* *Jan 2024 - Aug 2024*

- Assisted with recording session setup, including microphone placement, signal routing, and basic audio equipment configuration
- Provided technical and logistical support during live stage performances and events
- Studied music arrangement and audio recording techniques through hands-on observation and guided practice

## Projects

**Option Pricing using Quantum Computers** *Option-Pricing-on-Quantum-Computer*

- A from-scratch implementation of Stamatopoulos et al. (2020). "Option Pricing using Quantum Computers"

**A Guide to Quantum Computing** *A-Guide-to-Quantum-Computing*

- A collection of notes and implementations aimed at building intuition for quantum computing concepts without requiring prior knowledge of quantum mechanics.

**A Complete DNS Server** *DNS-Server-in-RUST*

- Developed a fully functioning DNS Server that handles DNS queries, resolves domain names, and returns IP addresses following standard DNS protocol specifications with Rust.

## Relevant Coursework

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| Thermodynamics and Statistical Mechanics | In Progress |
| Quantum Mechanics and Electrodynamics | In Progress |
| Intermediate Mechanics | In Progress |
| Partial Differential Equations | In Progress |
| Physics II Lab | In Progress |
| Foundations of Analysis I | A- |
| Intro to Probabilities | A |
| Intro to Quantum Computers - Graduate Course | A |
| Enhanced Intro to Differential Equations - Honors Course | B |
| Modern Physics | A |
| Survey of Numerical Analysis | A |
| Enhanced Linear Algebra - Honors Course | B+ |
| Foundations of Astronomy | A |
| Accelerated Object-Oriented Programming | A- |
| Physics I, II | A |
| Physics I Lab | A |
| Calculus I, II, III | A |